



MTH204 : Symmetry (2024-25)

Tutorial 02 (January 24, 2025)

1. Consider the dodecagon P_{12} and the group $\text{rot}(P_{12})$ of its rotational symmetries. Find all singleton subsets of $\text{rot}(P_{12})$ which are its generating sets.
2. Count the number of generating subsets of $\text{sym}(\Delta)$.
3. Write down the composition table of the group G that has the following properties.
 - (a) It has a generating set $S = \{a, b\}$.
 - (b) Relations in G are: $a^3 = 1, b^3 = 1, ab = ba$.
4. Draw the Cayley graph $\Gamma(\text{sym}(\Delta), S)$, for the following choices of S .
 - (a) $S = \{f_1, f_2\}$.
 - (b) $S = \{f_1, f_2, f_3\}$.
 - (c) $S = \{f, r\}$.
5. True or false? Justify. *"The group $\text{sym}(P_\infty)$ is the group of symmetries of a circle".*
6. True or false? Justify. *"If $f_1, f_2 \in \text{sym}(P_n)$ are two distinct reflections, then the set $\{f_1, f_2\}$ generates $\text{sym}(P_n)$ ".*
7. Find an infinite group G for which there is a generating set $\{a, b\} \subseteq G$ such that $a^2 = 1$ and $b^2 = 1$.
8. Find infinitely many groups G such that $\text{sym}(P_\infty) \subseteq G \subseteq \text{O}_2(\mathbb{R})$.
9. We know that $\text{sym}(P_\infty) \subseteq \text{O}_2(\mathbb{R})$. We also know that $\text{O}_2(\mathbb{R})$ does not contain translations. But, we also know that the translation $T_{(1,0)}$ belongs to $\text{sym}(P_\infty)$. Why isn't this a contradiction—if a subgroup contains a translation, how come the ambient group does not!
10. An element $g \in G$ is called *conjugate* to $h \in G$ if $g = xhx^{-1}$ for some $x \in G$. Show that being conjugate is an equivalence relation on G .
11. Show that the rotations R_α and $R_{-\alpha}$ are conjugate to each other in the group $\text{O}_2(\mathbb{R})$.
12. Are all reflections conjugate to each other in $\text{sym}(\Delta)$? What about $\text{sym}(\Delta)$?
13. If you are willing to venture into this! A subset of $\text{sym}(P_n)$ is picked uniformly randomly. What is the probability that this subset is a generating set of $\text{sym}(P_n)$? What are the upper and lower bounds of this probability? For which n are these bounds achieved?

14. Again, do this when you have enough time. Write down the composition table of the group G that has the following properties.

(a) It has a generating set $S = \{i, j\}$.

(b) Relations in G are: $i^4 = 1, iji = j, jij = i$
